

Matthew De Jesus

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EDUCATION

Massachusetts Institute of Technology (MIT)

Cambridge, MA

- Candidate for Masters in Engineering **Expected Graduation May 2027**
- **Relevant Coursework:** Digital Systems, Electrical Circuits, Power Electronics, Computation Structures, Microcomputers, Semiconductors, Nanoelectronics, Computer Systems Engineering, Fundamentals of Programming, Algorithms and Data Structures, Low-level Programming in C and Assembly

EXPERIENCE

MIT Research Laboratory of Electronics, Cambridge, MA

Jun 2025 - Present

Undergraduate Researcher

- Designed and iterated multiple high-frequency PCB test platforms to characterize switching loss, dynamic on-resistance and output capacitance losses in 5th generation GaN FETS.
- Automated laboratory instrumentation using SCPI (power supplies, oscilloscopes, thermal control) to execute voltage, frequency, and temperature sweeps for reproducible device characterization
- Co-authored paper accepted to the 2026 IEEE Applied Power Electronics Conference; to appear in IEEE Xplore

MIT Energy Initiative, Cambridge, MA

Jun 2024 - Aug 2024

Undergraduate Researcher

- Built a simulation to model energy arbitrage in the Texas power market, balancing speed and estimation accuracy
- Used a pruning approach to keep only the top N profitable paths each hour, reducing computation while focusing on the most profitable outcomes. Identified the highest-income scenario and generated clear revenue and ROI summaries.
- Added robust input checks and error handling for user-selected parameters (load zones, file types, etc.)

Bloomberg Summer Internship, New York, NY

Jun 2023 - Aug 2023

Intern, Employee Operations

- Managed computers on the Bloomberg network using Active Directory, including remote installation of programs
- Enrolled laptops into Microsoft Azure, ensuring seamless integration and enhanced functionality
- Led Scrum projects using the Agile framework, driving team collaboration and efficient project completion

PROJECTS

Differential Amplifier IC (Fabricated by Intel), Cambridge, MA

Feb 2025 - May 2025

- Designed a CMOS differential amplifier in Intel 22nm PDK achieving 80 dB gain, 1 mW power, 200 mV swing
- Implemented inverter, passgate, and digital code checker for integration into shared bus IC architecture
- Verified performance in Cadence (gain, swing, power, node voltages) and delivered DRC/LVS-clean submission
- Tuned transistor sizing and biasing to maximize figure of merit, balancing gain, swing, and power

Go-Cart Motor Drive, Cambridge, MA

Aug 2024 - Sept 2024

- Designed a PWM signal with adjustable duty cycle and frequency using a 555 timer and a 311 comparator
- Integrated delay circuitry to control two N-channel MOSFETs in a totem-pole configuration for efficient switching
- Achieved high-power handling capability with 36V and 30A, successfully driving a 1.5HP motor for a go-kart
- Demonstrated expertise in PWM motor control and power electronics, managing component selection and cost

RISC-V Processor, Cambridge, MA

Aug 2023 - Dec 2023

- Designed and implemented a 32-bit RISC-V 5-stage pipelined processor with a 600 ns critical path
- Integrated a 32-bit direct-mapped cache and a 2-way, 32-bit associative cache to enhance processor efficiency
- Optimized the processor by implementing branch prediction and full bypassing from writeback and execute

SKILLS

Programming Languages: Verilog, Python, C, Assembly, Javascript, Bluespec, LISP (Scheme), PSOC, FPGA

Electrical: Networking, Cable Routing, Soldering, Electronics Repair, Computer Building, Power Electronics